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***Abstract*—Most existing semantic communication systems process all semantic features equally during transmission, without considering their relative importance, resulting in inadequate transmission reliability of critical semantic features and consequently deteriorate overall system robustness and reconstruction quality. To address this limitation, in this paper, we propose a knowledge-base-assisted semantic communication system for wilderness search-and-rescue scenarios, where conventional wireless communication techniques are employed to enable differentiated transmission reliability of semantic features with different importance. To improve the quality of the reconstructed image, this paper designs a background image knowledge base, and Stable Diffusion Inpainting is adopted to generate background images for the knowledge base. To enhance transmission reliability and data rate, Adaptive Coding and Modulation (ACM) and diversity combining techniques, including maximal ratio combining (MRC), selection combining (SC), and majority voting combining (MVC), are incorporated into the system. We evaluate the effectiveness of the techniques through performance metric Bit Error Rate (BER), under Rayleigh fading channels with additive white Gaussian noise. Simulation results on the Search And Rescue Dataset (SARD) demonstrate that low-order modulation and low code rate decrease the BER of the cropped person images, and MRC achieves the best BER performance of the coordinates among the combining strategies. Furthermore, the proposed semantic communication system improves overall BER performance while maintaining high data rate and robust semantic reconstruction capability.**

***Index Terms*—Semantic communication, Adaptive Coding and Modulation, Diversity combining schemes, Bit Error Rate.**

I. Introduction

Semantic communication has recently gained significant attention as a promising technology, which has attracted considerable attention from both academia and industry [1]. Unlike conventional communication that aims at the accurate delivery of symbols, the semantic communication targets the precise content reconstruction with equivalent semantics. It has the potential to achieve ultra-low compression rates and extremely high transmission efficiency. Semantic communication is considered the future of mobile communication, aiming to address bandwidth limitations in modern high-capacity multimedia content transmission.

In the field of image semantic communication, existing deep learning-based research can generally be classified into two directions. The first direction relies on unimodal features, where neural network architectures are optimized to obtain high-quality semantic representations of images [2], [3]. In addition, noise-robust modules have been designed to improve the accuracy of image reconstruction under noisy channel conditions. The second direction emphasizes multimodal feature fusion, where textual and visual information are jointly exploited to achieve richer semantic representations [4]. Compared with unimodal approaches, multimodal models generally achieve better reconstruction performance, particularly in unstable network environments. Despite achieving promising results in certain scenarios, existing semantic communication models still face two critical challenges in complex human-centric scenarios: (1) the difficulty in accurately capturing the spatial position of the person in the image , which leads to distorted or unrealistic reconstructions; (2) semantic features are vulnerable to channel noise, which results in quality degradation during image transmission. In human-centric scenarios, such as wilderness search and rescue, accurate reconstruction of the coordinate and visual appearance of the target person is essential for reliable target identification and effective rescue coordination.

Therefore, to address the first challenge, we incorporate the bounding box coordinate of the target person as an auxiliary modality to complement visual features. Compared with textual descriptions, the bounding box coordinate provides an explicit representation of the spatial position in human-centric images. More generally, the bounding box coordinate can be viewed as a specific form of structured semantic keypoints. Therefore, the proposed framework is not limited to wilderness search and rescue scenarios and can be naturally extended to other visual scenarios by replacing the bounding box region at the specified coordinates in the background image with the cropped image of target person. These coordinates and cropped images provide explicit spatial constraints to guide semantic encoding and preserve structural consistency during transmission.

To address the second challenge, we introduce a background image knowledge base [5] and conventional wireless communication techniques to mitigate the impact of noise on semantic features. The background image knowledge base serves as a repository of the background images. By referencing a specific background image from the knowledge base, communication systems can more robustly infer and reconstruct the intended information, even when the transmitted data is corrupted or incomplete. Stable Diffusion Inpainting is adopted to generate the background images for this knowledge base. The background image knowledge base provides background information to recover the visual content of non-critical regions outside the target person, thereby making the reconstructed image more realistic. Furthermore, considering that different semantic features have different importance, conventional wireless communication techniques are incorporated to provide differentiated transmission reliability. Specifically, Adaptive Coding and Modulation (ACM) [6] and diversity combining techniques [7] are employed to improve transmission reliability and spectral efficiency.

Therefore, in this paper, we propose a knowledge-base-assisted semantic communication system for wilderness search-and-rescue scenarios. Specifically, we employ YOLOv8 to extract semantic features and reduce redundancy. By incorporating conventional wireless communication techniques and using the background image as additional auxiliary knowledge, we achieve superior image reconstruction performance, especially under challenging channel conditions. The major contributions of this paper are summarized as follows:

- We design a knowledge-base-assisted semantic communication system that enables the network to capture the spatial position of the person in the image often overlooked by prior approaches, thereby achieving higher-quality image reconstruction.
- We propose a novel background image knowledge base, which differs from prior knowledge base designs based on textual annotations or category-level semantic vectors. With background information through background images, the background image knowledge base provides background priors that directly complement visual semantics and significantly enhance decoding under challenging channel conditions.
- we employ conventional wireless communication techniques to mitigate the impact of noise on semantic features.

II. System Model

A. Semantic Encoder

The source image is encoded into two distinct semantic features [8], namely the bounding box coordinate of the target person and the cropped image of the target person from the source image, utilizing two semantic extractors [9] based on the YOLOv8 model and down sampling [10]. YOLOv8, proposed by Ultralytics in 2023 as an improved version of YOLOv5, is a one-stage object detection algorithm that integrates state-of-the-art techniques and introduces new functionalities. YOLOv8 not only enhances the accuracy of model but also increases computational speed. Therefore, YOLOv8 model is adopted as the semantic extractor in our system. The bounding box coordinate is extracted only using YOLOv8 model, the cropped image is extracted using YOLOv8 model with down sampling for further compression. For simplicity, we use subscripts 1 and 2 to replace subscripts coordinate and cropped image of person, respectively, in the following discussion. The ith extracted feature can be expressed by

$$\mathbf{S}_i = \mathcal{E}_i \left(\mathbf{X} \mid \boldsymbol{\theta}_i^* \right), \forall i \in \{1, 2\} \tag{1}$$

where $\mathcal{E}_i \left(\mathbf{X} \mid \boldsymbol{\theta}_i^* \right)$ is the ith model with $\boldsymbol{\theta}_i^*$ being network parameters and $\mathbf{X}$ is the source image.

To enhance domain adaptability, transfer learning [11] is employed by fine-tuning the pre-trained YOLOv8 weights using our wilderness search-and-rescue personnel dataset. This fine-tuning process enables the model to align more effectively with the semantic extraction task, thereby improving its robustness and localization accuracy in complex wilderness scenarios.

As reported in Table X, the YOLOv8 model fine-tuned on the wilderness search-and-rescue dataset shows a substantial improvement in detection accuracy, compared with the model using pre-trained weights without fine-tuning, thereby validating the effectiveness of the proposed transfer learning strategy.

To align with existing digital communication systems, the semantic feature $\mathbf{S}_i$ is transformed into a bit sequence $\mathbf{K}_i$, which can be expressed as

$$\mathbf{K}_i = \mathcal{B} \left(\mathbf{S}_i \right), \tag{2}$$

where $\mathcal{B} \left(\cdot \right)$ is a binary mapping function.

B. Transmission Module

Due to the different importance of the semantic data streams, multi-stream transmissions are considered in our system. The received data streams are expressed as

$$\left[\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2 \right] = \mathcal{T} \left(\left[\mathbf{K}_1, \mathbf{K}_2 \right] \right), \tag{3}$$

where $\mathcal{T} \left(\cdot \right)$ is the transmission scheme, which comprises diversity/combining, the channel coding/decoding and modulation/demodulation components.

In the transmission scheme, the bit sequences $\mathbf{K}_1$ and $\mathbf{K}_2$ are both encoded using LDPC channel coding, but with different code rates and modulation schemes [12]. The bit sequence $\mathbf{K}_1$ is transmitted using low-order modulation, low rate channel coding, and a diversity scheme, whereas $\mathbf{K}_2$ is conveyed via an adaptive coded modulation scheme.

Rayleigh fading channel with additive white Gaussian noise [13] is considered as the channel model in this paper to realistically simulate the wireless transmission environment in wilderness search-and-rescue scenarios.

C. Semantic Decoder

In the semantic decoder [14], the received semantic data streams $\hat{\mathbf{K}}_i$ are first reconverted to the semantic features $\hat{\mathbf{S}}_i$, which can be expressed by

$$\hat{\mathbf{S}}_i = \mathcal{B}^{-1} \left(\hat{\mathbf{K}}_i \right), \tag{4}$$

where $\mathcal{B}^{-1} \left(\cdot \right)$ is the inverse operation of $\mathcal{B} \left(\cdot \right)$. After obtaining the semantic features $\hat{\mathbf{S}}_i$, a background image $\mathbf{U}$ is required

TABLE I
CODING AND MODULATION SCHEMES SELECTED

	Coding Rate	Modulation Mode	R_m (bits/sym)
CMS1	$4/5$	BPSK	4/5
CMS2	$2/3$	BPSK	2/3
CMS3	$1/4$	16QAM	1
CMS4	$1/8$	16QAM	1/2
CMS5	$1/8$	64QAM	3/4
CMS6	$1/16$	64QAM	3/8

for further processing. The background image is provided from the knowledge base generated by stable diffusion inpainting model. Finally, the semantic features $\hat{\mathbf{S}}_i$ and the background image $\mathbf{U}$ are forwarded to the synthesizer D to synthesize $\hat{\mathbf{X}}$, which can be expressed as

$$\hat{\mathbf{X}} = D\left(\hat{\mathbf{S}}_1, \hat{\mathbf{S}}_2, \mathbf{U}\right) = \mathcal{D}\left(\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2, \mathbf{U}\right), \tag{5}$$

where $\mathcal{D}$ is the semantic decoder.

III. THE PROPOSED SEMANTIC COMMUNICATION SCHEME

A. Adaptive Coding and Modulation

In semantic communication systems, transmitted signals over noisy channels are still affected by various channel impairments, including path loss, shadowing, fading, and noise, which deteriorate the quality of the received signal. To overcome these impairments and enhance reliability while increasing system capacity, link adaptation techniques are employed to modify the transmitted signal before transmission according to the current channel state. One of the most widely used approaches is Adaptive Coding and Modulation (ACM). In ACM, the modulation order and coding rate are unfixed [15]. Lower-order modulation schemes and lower coding rates are selected when the channel is degraded, whereas higher-order modulation schemes and higher coding rates are used under favorable channel conditions to increase the data rate or to maintain certain transmission quality [16].

In this paper, the coordinates carry more critical semantic information. The reliability of the coordinates are more important than the cropped person image for the semantic decoder at the receiver. Therefore, the semantic data streams $\mathbf{K}_1$ can be transmitted using low-order modulation and low code rate, specifically 1/4 code rate and BPSK modulation. For the semantic data streams $\mathbf{K}_2$, we employ BPSK, 16-QAM, and 64-QAM modulations in combination with different code rates of LDPC coding. The code rates used are 4/5, 2/3, 1/4, 1/8, and 1/16. An adaptive coded modulation scheme used in this system is shown as Table I.

For the nth coded modulation scheme CMSn, if the code rate is R and M denotes the modulation order, the number of information bits carried by each symbol is $\mathrm{R_n} = \mathrm{R} \cdot \log_2 \mathrm{M}$.

B. Spatial Diversity and Combining

To ensure the reliability of the coordinates, the semantic data stream $\mathbf{K}_1$ employs the diversity and combining techniques, including Maximal Ratio Combining (MRC), Selection Combining (SC), and Majority Voting Combining (MVC).

Maximal Ratio Combining is a diversity technique designed to combat the effects of channel fading. At the receiver, L ($L \geq 2$) independently faded diversity signals are received and combined to reduce the impact of fading and improve the overall reception reliability.

Maximal Ratio Combining is a weighted combining technique [17] in which multiple received signals are combined using channel state information (CSI) to maximize the signal-to-noise ratio (SNR), and the corresponding expression is given as follows:

$$y_{\mathrm{MRC}} = \frac{\sum_{i=1}^{L} h_i^* y_i}{\sum_{i=1}^{L} |h_i|^2}, \tag{6}$$

Where h_i represents the channel fading coefficient of the ith branch, y_i represents the received signal of the ith branch, and L denotes the number of receiving branches [18]. In this study, the number of receiving branches is set to 3, 6 and 9 for performance comparison.

SC selects the single branch with the highest channel gain [19], avoiding complex computations by simply comparing branch metrics. While its low-complexity design is resource-efficient, SC does not fully use multi-branch diversity, leading to slightly degraded performance compared to MRC. The corresponding expressions are given as follows:

$$k = \arg\max_{1 \leq i \leq L} |h_i|^2 \tag{7}$$

$$y_{\mathrm{SC}} = y_k \tag{8}$$

Where h_i represents the channel fading coefficient of the ith branch, y_i represents the received signal of the ith branch, L denotes the number of receiving branches, and k denotes the index of the receiving branch with the maximum channel gain.

In MVC, the receiver compares the bit from all branches and selects the bit that appears most frequently.

C. Background Image Knowledge Base

With the advancement of knowledge engineering, an increasing number of knowledge-based systems have been developed and used in various domains.

The proposed background image knowledge base is applied to the reconstruction of target person images in our wilderness search-and-rescue task. The background image knowledge base provide a background image as input to the synthesizer.

Inpainting is the task of filling masked regions of an image with new content either because parts of the image are corrupted or to replace existing but undesired content within the image.

Stable Diffusion Inpainting is a latent text-to-image diffusion model [20] that can generate photo-realistic images from text input and can also perform inpainting by using a mask.

In this paper, the Stable Diffusion Inpainting is used to generate a background image $\mathbf{U}$ as the background image knowledge base. To improve its performance, several improvements are applied to the model inputs. The Stable Diffusion Inpainting model requires three inputs, including a text prompt, an input image, and a corresponding mask image. The mask image is obtained by directly processing the input image with a pretrained background removal model, namely rembg.

IV. Simulation Results and Analysis

Search And Rescue Dataset (SARD) was used to train and evaluate the performance of the proposed semantic communication system. For the SARD, with a resolution of 640 × 640, training and testing were performed on Rayleigh fading channel with additive white Gaussian noise. The proposed semantic communication system is evaluated over the channel with an SNR uniformly distributed between 0 dB and 16 dB.

To transmit the coordinate and cropped person image data streams, the communication parameters are set as follows. The modulation and coding schemes of these two semantic data streams are different. The semantic data stream $\mathbf{K}_1$ employs the diversity and combining techniques, whereas $\mathbf{K}_2$ does not.

As shown in Table I, MCSn is selected as the coded modulation scheme for the cropped person image data streams. The coordinate data streams employ LDPC coding with a rate of 1/4 and BPSK modulation. Several simulations are done to evaluate the BER performance of both the cropped person image data streams and the coordinate data streams, as well as the overall BER. Note that these simulations are done over 269 cropped person images and coordinates of them using Monte Carlo simulation approach. Fig. 1 shows

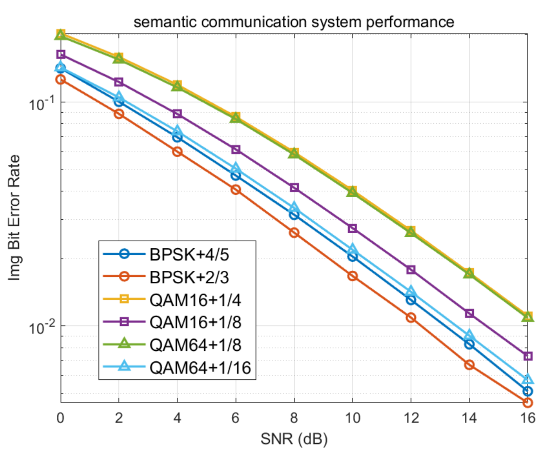

Fig. 1. The BER performance of the cropped person image under different coded modulation schemes.

the BER performance of the cropped person image under six coded modulation schemes. The discrete points represent the simulated points, while the solid lines are fitted lines. The results given in Fig. 1 show that the image bit error rate decreases as the channel coding rate decreases. In addition, decreasing the modulation order also contributes to a decrease in the image bit error rate. The total bit error rate results are given in Fig. 5, demonstrating the similar performance lines to the image bit error rate.

Most prior semantic communication systems do not use diversity and combining techniques in transmission. However, we employ diversity and combining techniques to the semantic communication system and analyze its performance.

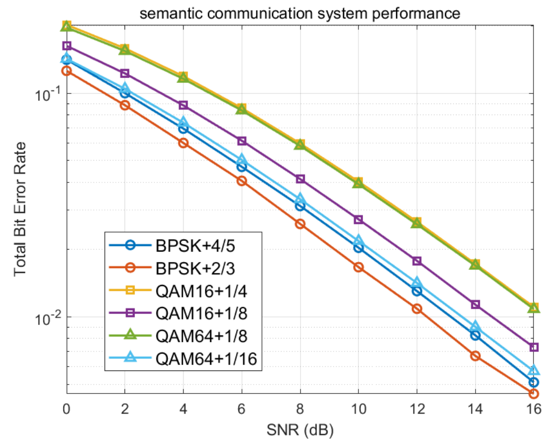

Fig. 2. The BER performance of the total under different coded modulation schemes.

Because the coordinate data streams are more important for the system, diversity and combining techniques are applied to the coordinate data streams, whereas they are not employed for the cropped person image data streams. Fig. 3 shows the BER performance of the coordinate under six different merging strategies. The curves presented consider the BER performance under three combining techniques, MRC, SC, and MVC at different number of receiving branches with $L = 3$ and $L = 9$. The coordinate bit error rate decreases as the number of receiving branches L increases. It can be observed that, for the same number of receiving branches L, MRC achieves the best BER performance, followed by SC and then MVC. The MRC approach can fully utilize the diversity gain. Fig. 4 shows the BER performance of the cropped person image under six different merging strategies. Fig. **??** shows the total bit error rate results under six different merging strategies. The two BER performance are very similar. Because diversity and combining techniques are not applied to the cropped person image data stream, the total and image bit error rate results under the six different combining strategies are highly consistent. This suggests that diversity and combining techniques better preserves important semantic information transmission of images in scenarios with significant noise interference, thereby decreasing the coordinate bit error rate of images.

V. Conclusion

In this paper, we proposed a novel semantic communication system based on knowledge base, characterized by the fact that the system transmits only the critical semantic information, while the remaining non-essential information is recovered at the receiver based on prior knowledge stored in a knowledge base, simplifying the communication process and reducing storage requirements. By introducing ACM, diversity and combining techniques, our system effectively improves BER performance by transmitting semantic information at higher data rate and certain transmission quality, while decreasing the coordinate bit error rate of images, thereby improving image

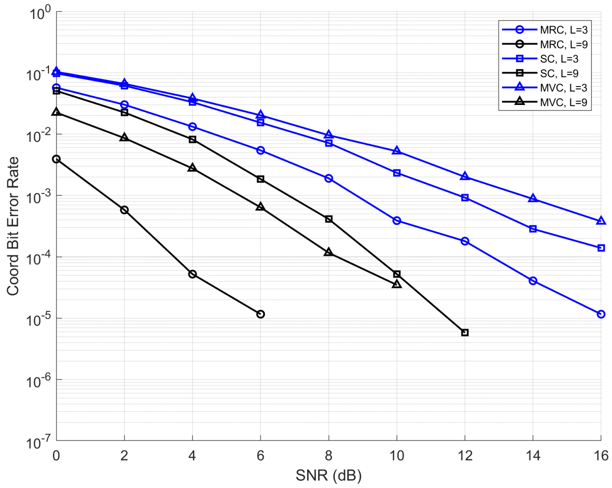

Fig. 3. The BER performance of the coordinates under different merging strategies.

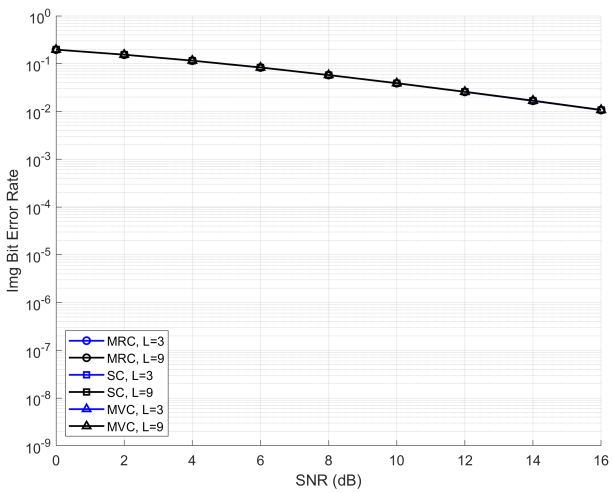

Fig. 4. The BER performance of the cropped person image under different merging strategies.

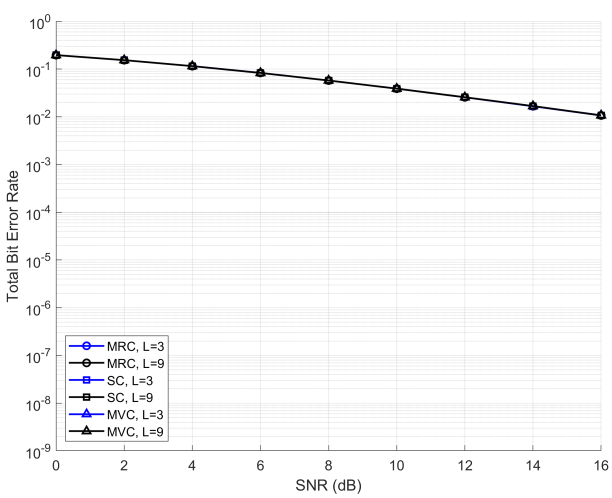

Fig. 5. The BER performance of the total under different merging strategies.

quality. Simulation results demonstrate that the semantic communication system employing the proposed method achieves performance improvements.

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